

Question	Answer	Marks
7(a)	circle(s) drawn only around β^- and $\bar{\nu}$ symbols	B1
7(b)	(electron) <u>antineutrino</u>	B1
7(c)	kinetic (energy)	B1
7(d)	Y has one more proton (and one less neutron)/X has one less proton (and one more neutron) or Y has more protons (and fewer neutrons)/X has fewer protons (and more neutrons) or a neutron changes to a proton or the number of protons increases	M1
	(so) not isotopes	A1
7(e)	up down down changes to up up down or $udd \rightarrow uud$ or down changes to up or $d \rightarrow u$	B1